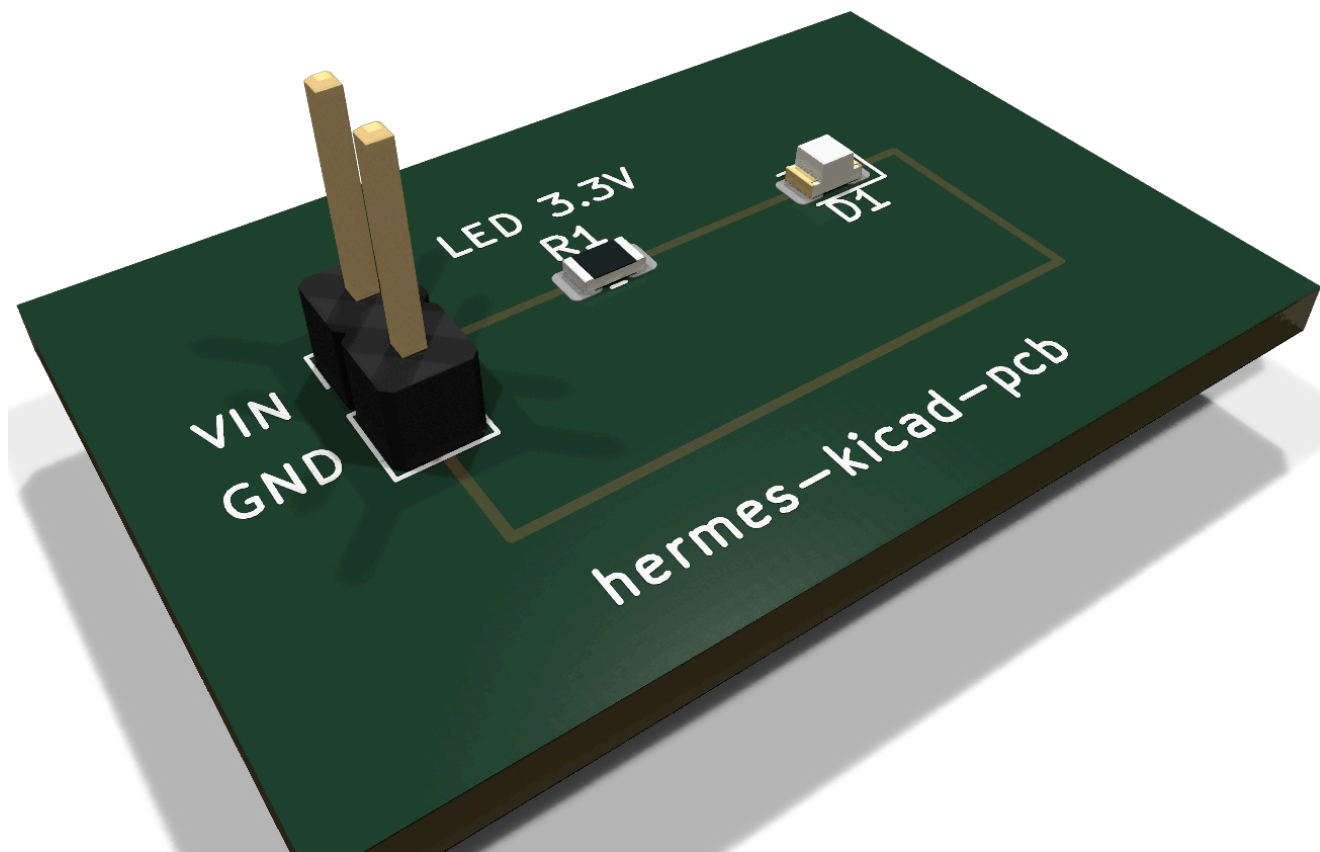


Simple LED board

Hardware design document

Revision A · 2026-09-22 · DEMONSTRATION DESIGN



Verification at document generation

ERC	PASS (0 open)
DRC	PASS (0 open)
Connectivity	0 unconnected
Schematic <-> board parity	0 differences
Reviewed exceptions	0 (justified in section 6)

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Document control

Item	Value
Title	Simple LED board
Revision / status	A / DEMONSTRATION DESIGN
Generated	2026-09-22 by the kicad-pcb skill (design_doc.py)
Tools	KiCad 10.0.6 (kicad-cli)

Source files

File	SHA-256 (first 16)
simple-led.kicad_sch	25db7eaf04f50880
simple-led.kicad_pcb	8a918edc70e15ba4

The hashes tie this document to the exact design files it was generated from. Regenerate the document after any change.

Contents

1. Design description and theory of operation
2. 3D schematic
3. Schematic
4. Printed circuit board
5. Bill of materials
6. Verification
7. Manufacturing outputs
8. Open issues and limitations

1. Design description and theory of operation

Purpose and scope

The smallest complete board the kicad-pcb skill produces with nothing but KiCad installed (no Freerouting, no Java): a 2-pin power header, a series resistor and an LED. It exercises schematic generation, board generation linked to the schematic, deterministic routing, ERC, DRC with schematic parity and the fabrication outputs.

Requirements

ID	Requirement	Source	Status
R1	Light one LED from an external DC supply V_{IN}	demo scope	implemented
R2	LED current set by the generator options	--vin, --led-vf, --led-ma	implemented (E12 value)
R3	Single-layer routing, 30 x 20 mm board	demo scope	implemented

Theory of operation

The supply enters on J1 (pin 1 = VIN, pin 2 = GND). R1 limits the LED current: $R = (V_{IN} - V_f) / I_{LED}$, rounded **up** to the next E12 value so the current never exceeds the target. D1's anode (pad 2) takes the resistor side, its cathode (pad 1) returns to GND.

Design calculations

Case	Calculation	R1 (E12)	Resulting current
Default: 5 V, Vf 2.0 V, 3 mA	$(5 - 2.0) / 3 \text{ mA} = 1.0 \text{ k}\Omega$	1 k Ω	3.0 mA
3.3 V, Vf 2.0 V, 2 mA	$(3.3 - 2.0) / 2 \text{ mA} = 650 \Omega$	680 Ω	1.9 mA

Resistor dissipation at the default: $(3.0 \text{ V})^2 / 1 \text{ k}\Omega = 9 \text{ mW}$ — far below the 125 mW standard rating of an 0805 resistor.

PCB layout notes

All parts on the top layer, one 0.3 mm track per net, GND returned along the bottom edge of the board. Silkscreen marks VIN/GND next to the header pins.

Assumptions and unverified data

- LED forward voltage (default 2.0 V) is an input, not a measured value; the exact LED part number is TBD.
- No reverse-polarity protection: a reversed supply only reverse-biases the LED, which most LEDs tolerate only up to about 5 V (check the chosen LED's datasheet).

Open issues

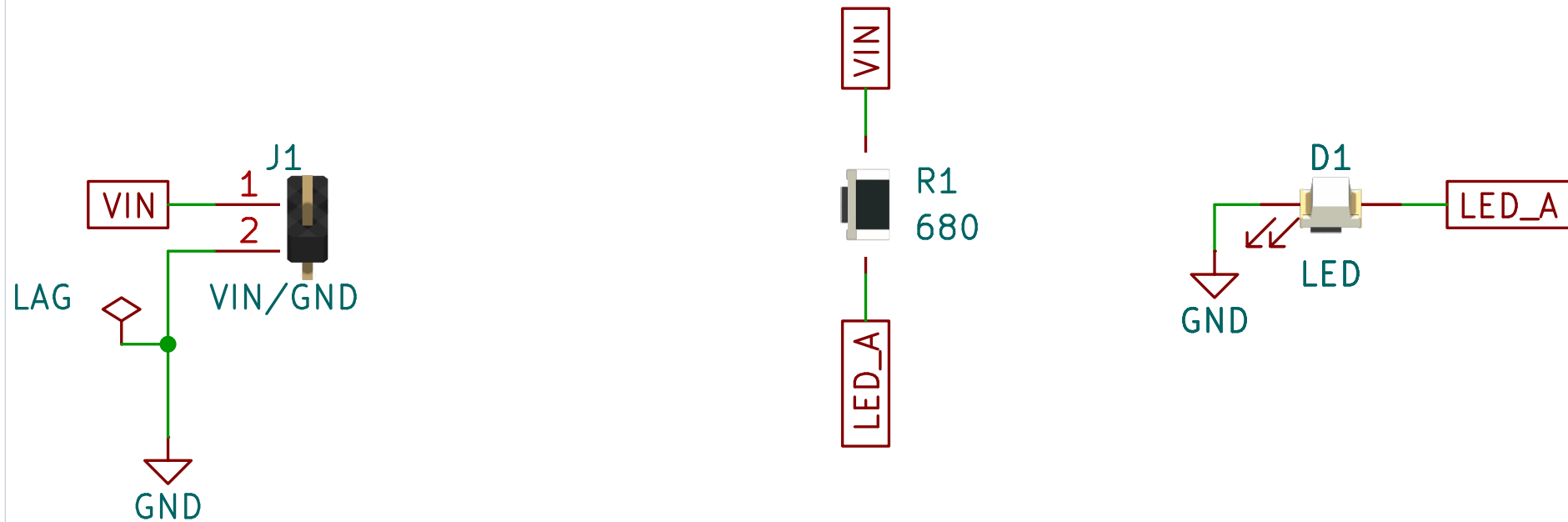
1. Choose exact LED and resistor part numbers before ordering.
2. Add reverse-polarity protection if the supply can be connected backwards above 5 V.

Design facts (extracted from the netlist)

Item	Value
Components	3
Nets	3
Sheets	1
Supply / ground nets	GND, VIN

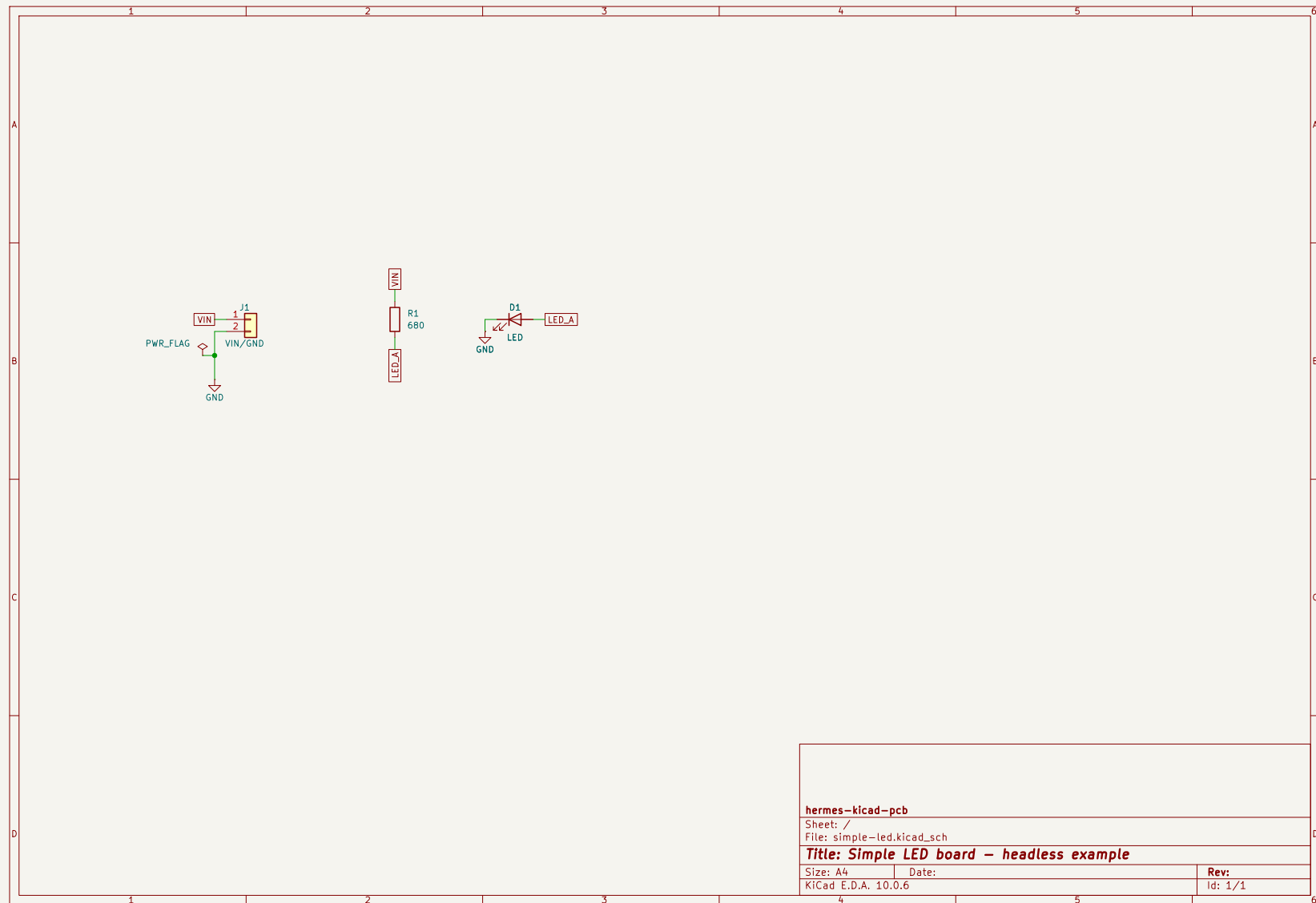


2. 3D schematic · simple-led



Parts drawn with their footprint's 3D model (KiCad renders). Wiring, labels and power symbols are the schematic itself.

3. Schematic

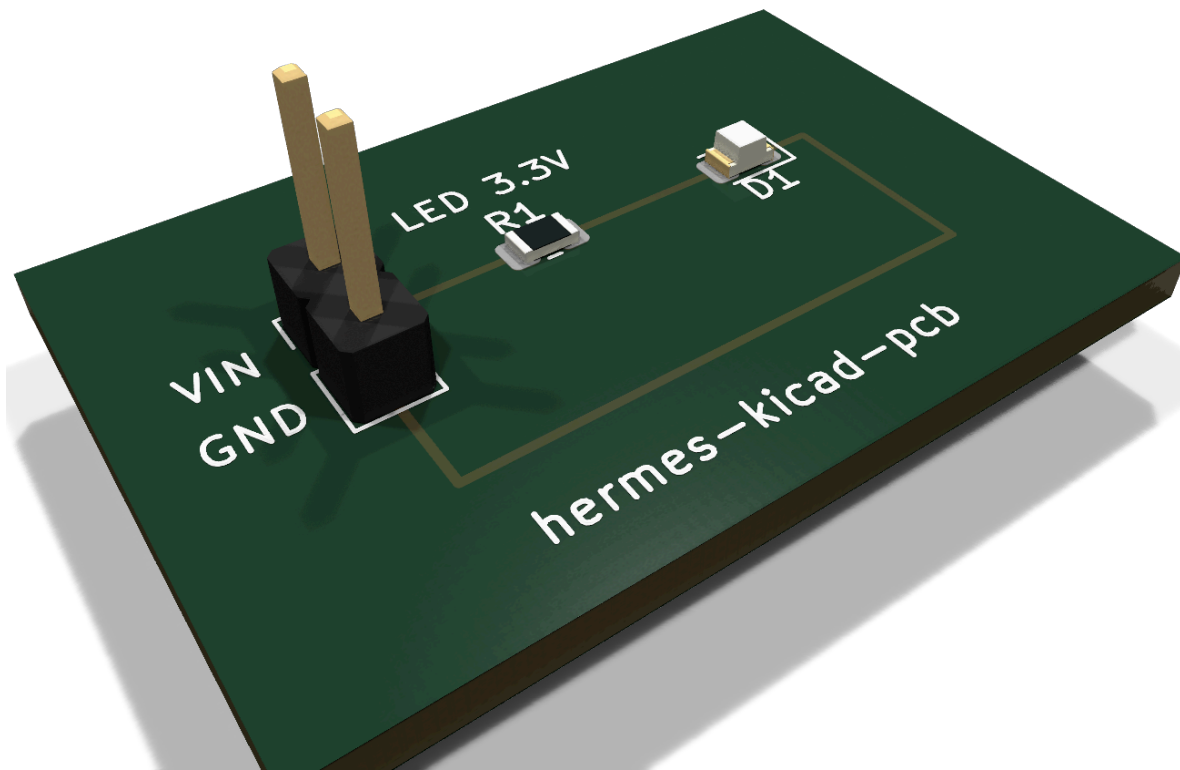


4. Printed circuit board

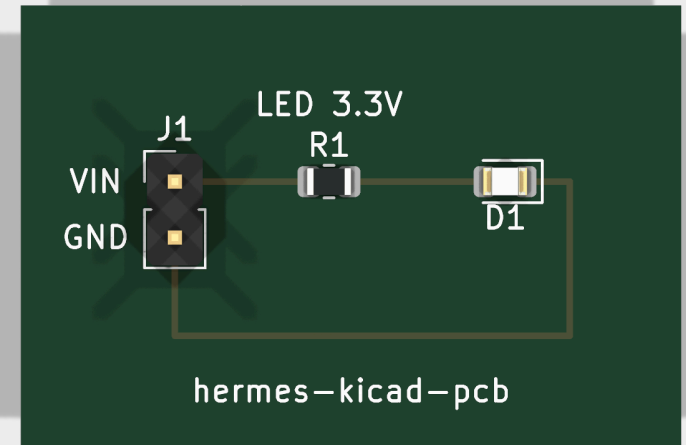
Property	Value
Outline	30.00 x 20.00 mm
Copper layers	2 (F.Cu, B.Cu)
Board thickness	1.6 mm
Footprints	3
Track segments / arcs / vias / zones	6 / 0 / 0 / 0
Narrowest track	0.3 mm

Board views and vector layer plots follow on the next pages (KiCad raytraced renders with the footprints' own 3D models).

4. Board views (KiCad 3D renders)



3D view

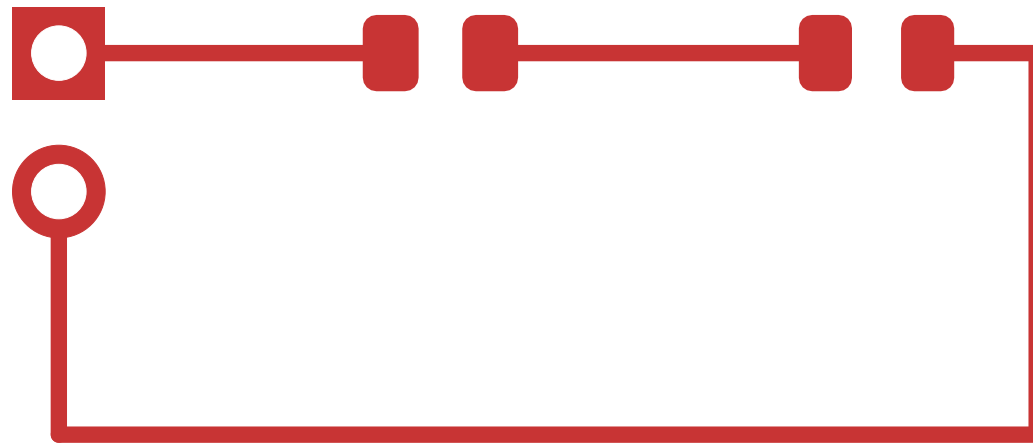


Top side



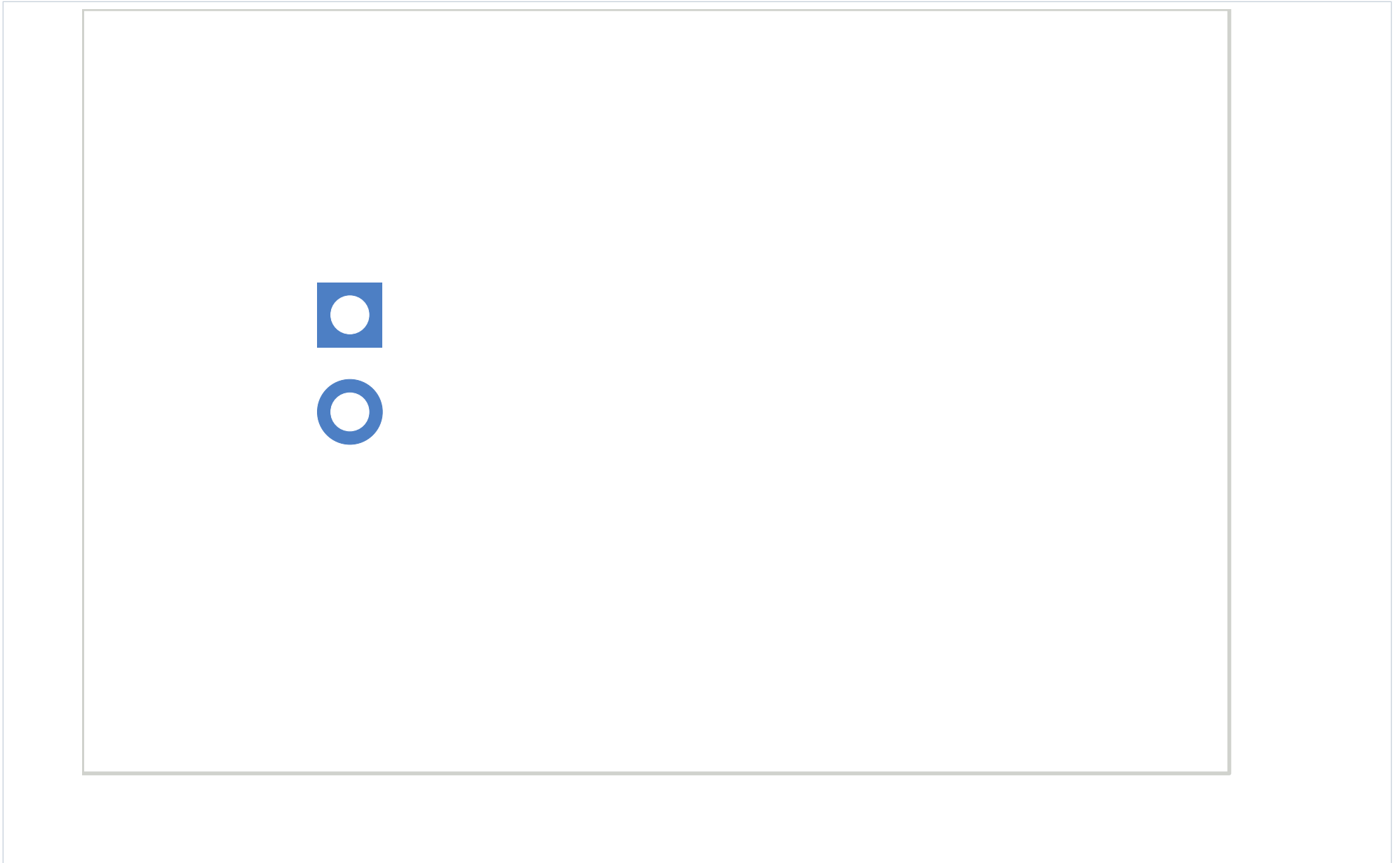
Bottom side

4. Layer plot · F.Cu



Vector plot from kicad-cli (board outline on every page); zoomed to the board.

4. Layer plot · B.Cu



Vector plot from kicad-cli (board outline on every page); zoomed to the board.

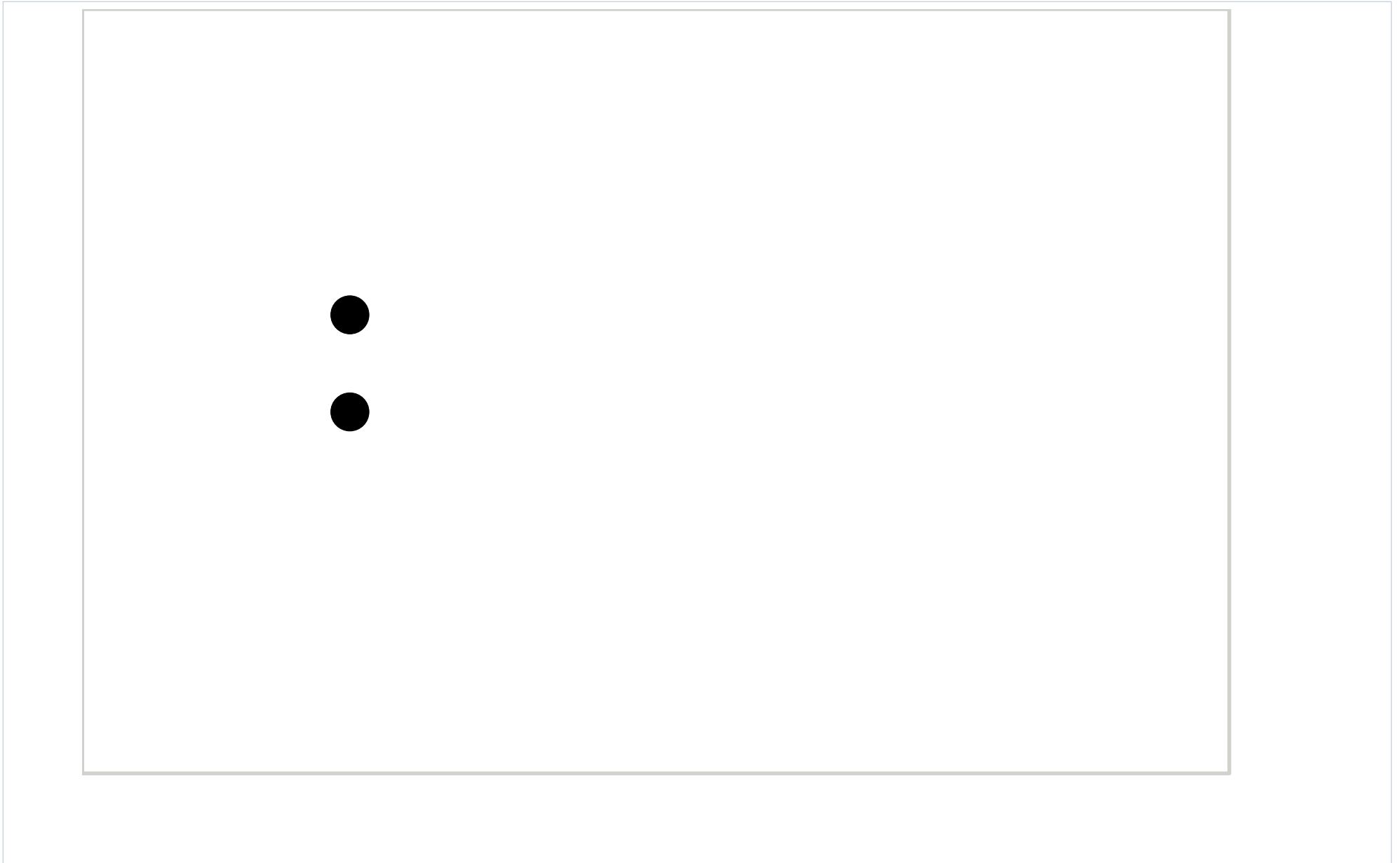
4. Layer plot · F.Silkscreen



hermes-kicad-pcb

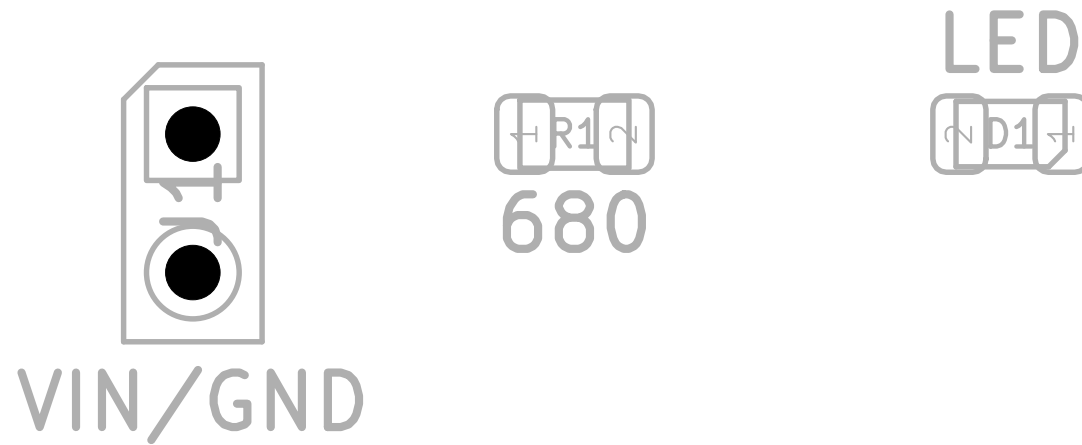
Vector plot from kicad-cli (board outline on every page); zoomed to the board.

4. Layer plot · B.Silkscreen



Vector plot from kicad-cli (board outline on every page); zoomed to the board.

4. Layer plot · F.Fab



Vector plot from kicad-cli (board outline on every page); zoomed to the board.

5. Bill of materials

Grouped from the schematic netlist. Supplier columns appear only when the schematic carries the field (LCSC / MPN); no part number or stock class is inferred.

Qty	References	Value	Footprint	Description
1	D1	LED	LED_0805_2012Metric	Light emitting diode
1	J1	VIN/ GND	PinHeader_1x02_P2.54mm_Vertical	Generic connector, single row, 01x02, script generated (kicad-library-
1	R1	680	R_0805_2012Metric	Resistor

6. Verification

Check	Result
ERC	PASS (0 open)
DRC	PASS (0 open)
Connectivity	0 unconnected
Schematic <-> board parity	0 differences
Reviewed exceptions	0 (justified in section 6)

ERC (all severities)

No open ERC findings.

DRC (all severities)

No open DRC findings.

Unconnected items

None.

Schematic parity

Board matches the schematic.

Reviewed exceptions

None.

ERC/DRC measure rule consistency against the project's design rules. They do not replace the engineering review of power, RF, thermal and mechanical behaviour listed in section 8.

7. Manufacturing outputs

Fabrication archive `simple-led-gerbers.zip` (SHA-256 `f62b22fc81f9d0fd`), 11 files:

- `simple-led-B_Cu.gbl`
- `simple-led-B_Mask.gbs`
- `simple-led-B_Paste.gbp`
- `simple-led-B_Silkscreen.gbo`
- `simple-led-Edge_Cuts.gml`
- `simple-led-F_Cu.gtl`
- `simple-led-F_Mask.gts`
- `simple-led-F_Paste.gtp`
- `simple-led-F_Silkscreen.gto`
- `simple-led-job.gbrjob`
- `simple-led.drl`

Check every output against the selected fabricator's current capabilities ([references/jlcpcb-rules.md](#) for the JLCPCB class) before ordering.

8. Open issues and limitations

1. Choose exact LED and resistor part numbers before ordering.
2. Add reverse-polarity protection if the supply can be connected backwards above 5 V.

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